Tie Zheng, Ph.D.

Postdoctoral Researcher, Department of Earth System Science, Stanford University | Stanford, CA
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Google Scholar | LinkedIn | ResearchGate

Research focus: Building electrification and environmental health; Exposure science; Air pollution modeling and measurements; Emission factor characterization; Instrumentation and field campaigns

I combine multi-platform monitoring with data-driven models to quantify air pollution dynamics and human exposure in homes and cities. My goal is to inform building electrification strategies by evaluating their exposure and health benefits.

Outside research, I am a competitive distance runner and part-time coach. I placed 6th overall in the 2026 San Francisco Marathon (fastest Chinese runner).

Education

Tsinghua University — Ph.D. | 2021–2025 (conferred 06/19/2025) | Advisor: **Ye Wu** | GPA 3.82/4.0
Major in Environmental Science and Engineering; Minor in Big Data Project
Dissertation: *Fine-Grained Mapping of Air Pollution Using Mobile Monitoring and Machine Learning*

Shanghai Jiao Tong University — M.Eng., Transportation Engineering | 2021 | Advisor: **ZR Peng** | GPA 3.68/4.0
Dissertation: *Spatiotemporal Distributions of Traffic-related Pollutants in Urban Road Environments*

Chongqing University — B.Eng., Civil Engineering | 2018 | Advisor: **Zhitao Yan** | GPA 3.50/4.0

Research Experience

Stanford University — Postdoctoral Researcher, 08/2025–present | Advisor: **Rob Jackson** | Earth System Science
- ❖ Led Ultrafine Particle (UFP) project in Electrification for Health program: We report, to our knowledge, the first estimates of size-resolved UFP emissions from gas stoves across multiple homes.
- ❖ Performed year-long UFP emission measurements in 60+ San Francisco Bay Area homes; derived energy-normalized, size-resolved UFP emission factors from residential gas appliances.

Tsinghua University — Doctoral Researcher, 2021–2025 | Advisor: **Ye Wu** | School of Environment
- ❖ Mapped hyperlocal air pollution in cities using machine learning and large-scale mobile measurement datasets (~1,500 h). Led multi-platform street-by-street mobile monitoring.
- ❖ Field campaign lead, NSFC-funded project: Hyperlocal air pollution field campaign in Macau.

Spatial Data Lab, Harvard University — Research Intern, 03/2025–09/2025 | Advisor: **Siqin Wang**
- ❖ Integrated large geospatial datasets with statistical and machine-learning methods for exposure mapping.

MIT — Visiting Researcher, 07/2024–12/2024 | Advisors: **Carlo Ratti, Simone Mora** | Urban Planning
- ❖ Data analyst for the City Scanner project: turning vehicles into sensing platforms.

Imperial College London — Visiting Researcher, 02/2024–06/2024 | Advisor: **Audrey de Nazelle**
- ❖ Worked on cost-effective mapping of hyperlocal air pollution, linking air pollution and human exposure.

Macau University of Science and Technology — Visiting Researcher, 01/2024–02/2024 | Advisor: **You Yan**
- ❖ Conducted field investigation and data collection for hyperlocal air quality in Macau, China.

Chinese Academy of Environmental Planning — Research Intern, 05/2022–08/2022 | Advisor: **Bofeng Cai**
- ❖ Worked on innovative technology for carbon peaking and carbon neutrality in China.

Publications

** Corresponding author; # equal contribution.*

Manuscripts in development or under review

[1] **Zheng, T.**, Derrick, S., Finnegan, C. J., Pongetti, J., Jourdain, C., Lebel, E. D., Kashtan, Y. S., Lucha, N., Huang, G., Michanowicz, D. R., Shonkoff, S. B. C., Boies, A. M., & Jackson, R. B. (2026). Ultrafine particle emissions from burning gas in homes. *In Development*.

[2] **Zheng, T.**, Wen, Y., Xiang, S., et al. (2026). A cost-effective framework for urban air pollution mapping. *Under Review (first revision), ACS ES&T Air.*

[3] Sumaiya P. K., **Zheng, T.***, Wen, Y., Xiang, S., et al. (2026). Multi-source domain adaptation for mobile-to-ambient air-pollution mapping. *In Development*.

Peer-reviewed journal articles

[1] **Zheng, T.**, Wen, Y., Xiang, S., Yang, P., Zheng, X., You, Y., Zhang, S., & Wu, Y. (2025). Cost-effective mapping of hyperlocal air pollution using large-scale mobile monitoring and land-use machine learning. *ACS ES&T Air*, 2(2), 151–161. **Journal cover.**

[2] Xiang, S., Yu, J., Yu, Y. T., et al., including **Zheng, T.** (2025). Exploring the holiday effect on elevated traffic-related air pollution with hyperlocal measurements in Chengdu, China. *Atmosphere*, 16(2), 171.

[3] **Zheng, T.**, Xiang, S., Zhang, S., & Wu, Y. (2023). Variability of traffic-related air pollutants at two- and four-phase intersections. *Atmospheric Pollution Research*, 14(12), 101936.

[4] **Zheng, T.**, Peng, Z. R., He, H. D., Zhang, S. J., & Wu, Y. (2022). Horizontal profiles of size-segregated particle number concentration and black carbon beside a major roadway. *Atmospheric Environment: X*, 16, 100187.

[5] **Zheng, T.#**, Jia, Y. P.#, Zhang, S., Li, X. B., Wu, Y., Wu, C. L., He, H. D., & Peng, Z. R. (2021). Impacts of vegetation on particle concentrations in roadside environments. *Environmental Pollution*, 282, 117067.

[6] **Zheng, T.**, Wang, H. W., Li, X. B., Peng, Z. R., & He, H. D. (2021). Impacts of traffic on roadside particle variations in varied temporal scales. *Atmospheric Environment*, 253, 118354.

[7] **Zheng, T.**, Li, B., Li, X. B., Wang, Z., Li, S. Y., & Peng, Z. R. (2021). Vertical and horizontal distributions of traffic-related pollutants beside an urban arterial road based on unmanned aerial vehicle observations. *Building and Environment*, 187, 107401.

[8] Jia, Y. P., Lu, K. F., **Zheng, T.**, Li, X. B., Liu, X., Peng, Z. R., & He, H. D. (2021). Effects of roadside green infrastructure on particle exposure: A focus on cyclists and pedestrians on pathways between urban roads and vegetative barriers. *Atmospheric Pollution Research*, 12(3), 1–12.

[9] Gao, Y., Wang, Z., Li, C., **Zheng, T.**, & Peng, Z. R. (2021). Assessing neighborhood variations in ozone and $PM_{2.5}$ concentrations using decision tree method. *Building and Environment*, 188, 107479.

[10] Wang, J., Yin, J., Khan, R. U., Wang, S., & **Zheng, T.** (2021). A study of inbound logistics mode based on JIT production in cruise ship construction. *Sustainability*, 13(3), 1588.

Book chapters

[11] Yu, Y. T., Zhang, S., Xiang, S., Yang, P., Wang, H., **Zheng, T.**, & Wu, Y. (2025). The role of ground-based mobile monitoring in air quality evaluation. In J. R. Barker, A. L. Steiner, & T. J. Wallington (Eds.), *Urban and regional air quality: Emissions, chemistry, and impacts* (pp. 267–323). World Scientific Publishing.

[12] Zhang, L., Cai, B., & **Zheng, T.** (2023). Developing an innovative technology system for carbon peaking and carbon neutrality. In J. Wang & H. Xu (Eds.), *Introduction to carbon dioxide peaking and carbon neutrality* (pp. 345–361). China Science and Technology Press. [In Chinese]

Presentations

- Stanford Global and Planetary Health Research Convening, 12th Annual (2026)
- International Symposium on Spatiotemporal Data Science (2026)
- Transportation Research Board, 104th Annual Meeting (2025)
- 7th International Students' Ecology and Environment Forum (2024) **Second Prize in Oral Presentation**
- UK and Ireland Exposure Science Meeting (2024)
- AGU24 Annual Meeting (2024)
- Urban Transitions (2024)
- 25th International Transport and Air Pollution Conference (2023)
- 16th and 17th National Environmental Conference for Doctoral Students, Tsinghua University (2022–2023)
- Transportation Research Board, 99th–102nd Annual Meetings (2020–2023)

Patent, Editorial Service, and Research Grants

- Co-inventor, "Air Quality Prediction System Based on Temporal Gate Mechanism," Chinese patent ZL 202310185638.6
- Guest Editor, *Atmosphere*, 2026–2027 Special Issue "Air Pollution Monitoring, Traffic-Related Emissions, and Health Exposure Assessment," Air Quality section; submissions open through March 2027
- Visiting Research Fund, Tsinghua University, 2024
- Conference Travel and Registration Support, TAP 2023, Gothenburg, Sweden, 2023
- Undergraduate Research Fund, *Wind-Induced Deflection of Suspension Insulators and Mitigation Technologies*, Chongqing University, 2017

Honors and Awards

- Three-time Tsinghua Comprehensive Scholarship, 2023–2025
- Tsinghua University Hash Scholarship, 2024
- Outstanding Team Award (Team leader), Big Data Project Practice Course, 2023
- Top 10 Athlete of the Year, Tsinghua University, 2023
- Outstanding Intern, Chinese Academy of Environmental Planning, 2022

Mentoring, Teaching and Leadership

- **Mentor**, Stanford SURGE Program, Doerr School of Sustainability, 2026
  Mentee: Sumaiya P. K., Gettysburg College
  Working Topic: *Mapping Air Health Inequity with AI and Community-Based Sensing*
- Stanford Postdoc Mentoring Certificate, 2026
- Selected Participant, Stanford Preparing Future Teaching Professors Program, 2026
- Running Coach, Palo Alto, CA, 2026–present
- Teaching Assistant, Tsinghua University, 2023–2025
- Team leader, Big Data Project Practice Course, Tsinghua University, 2023
- Vice-Captain, Tsinghua University Running Team, 2023

Peer Review

- **Reviewer for 23 academic journals**, 2021–present
- **Atmospheric science and air quality (4):** Air; Atmosphere; Atmospheric Environment; Atmospheric Environment: X
- **Environmental science, sustainability, and public health (7):** International Journal of Phytoremediation; Journal of Hazardous Materials; Pollutants; Science of the Total Environment; Sustainability; Toxics; Trends in Public Health
- **Buildings, cities, and land use (4):** Building and Environment; Land; Smart Cities; Urban Science
- **Transportation and energy (3):** Journal of Traffic and Transportation Engineering (English Edition); Transportation Research Part D: Transport and Environment; World Electric Vehicle Journal
- **Computing, engineering, and multidisciplinary science (5):** Algorithms; Applied Sciences; Automation; Earth Science Informatics; Scientific Reports

Technical Skills

- **Measurement and Data Collection:** Multi-pollutant high-resolution monitoring; mobile and UAV-based monitoring; instrumentation and platform development
- **Exposure and Modeling:** Exposure characterization; inhaled-intake estimation; multizone modeling; land-use regression; machine learning; spatiotemporal statistics; uncertainty and sensitivity analysis
- **Computing:** Python (advanced), R, SQL, ArcGIS Pro, LaTeX
- **Languages:** Mandarin (native) and English (fluent)

Athletic Achievements

Personal bests: 10K 30:14; half marathon 1:08:37; marathon 2:27:01

- 6th overall, San Francisco Marathon (Full), 2026
- 5th, San Jose Half Marathon, 2026
- 3rd, Rock ‘n’ Roll San Jose 10K (road race), 2025
- 1st (**meet record**), Men’s 10,000m, Beijing Collegiate Track and Field Championships, 2022, 2024
- 1st (**meet record**), Men’s 5,000m, London University Outdoor Championships, 2024
- 1st (**meet record**), Men’s marathon, Huai Rou Great Wall Marathon, 2023
- 1st overall, Victoria Park Half Marathon, London, 2024
- 1st, Men’s 5K (road race), Boston Mile for Migraine Race, 2024
- 1st, Men’s 100m Backstroke, Tsinghua University Swimming Competition, 2022
- 1st (**meet record**), Men’s 800m, 1,500m, 5,000m, Tsinghua University Track and Field Championships, 2022, 2023, 2024